Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Surface Temperature Thematic Assembly Centre For Global Ocean GMPE Sea Surface Temperature Multi Product Ensemble

SST_GLO_PHY_L4_NRT_010_005

Issue: 3.0

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CHANGE RECORD

Issue	Date	§	Description of Change	Author	Validated By
1.0	15/01/2014	all	First version of document	A. McLaren et al.	
1.1	31/03/2014	all	Minor re-wording following initial review	A. McLaren	
1.2	15/09/2014	all	Changes following review	A.McLaren et al.	
1.3	1/9/2015	all	Change format to fit CMEMS graphical rules. Introduce references to CMEMS.	B. Hackett	L. Crosnier
1.4	19/9/2017		Update to add new analyses to the ensemble	S. Good, V. Hernaman	
2.0	15/01/2021	all	Template update	S.Good	Bruno Buongiorno Nardelli
2.1	20/05/2021	I	Plots added and table updated.	S. Good	
NA	31/12/2022	all	Product removed from Copernicus Marine Service catalogue follwoing decision made after UK exiting EU	NA	
3.0	12/11/2025 2025	All	Creation of the document based on CMEMS-OSI-QUID-010-005-V2.1 as product re-enter Copernicus Marine Service catalogue and update to add new analyses to the ensemble	C. Mao, S. Good	A. Pisano

ACRONYM TABLE

Acronym	Description
AVHRR	Advanced Very High-Resolution Radiometer
CMC	Canadian Meteorological Center
DMI	Danish Meteorological Institute
FNMOC	Fleet Numerical Meteorology and Oceanography Centre
GAMSSA	Global Australian Multi-Sensor SST Analysis
GHRSST	Group for High-Resolution Sea Surface Temperature
GMPE	GHRSST Multi Product Ensemble
G1SST	Global 1 km Sea Surface Temperature
JPL	Jet Propulsion Laboratory
MGDSST	Merged Satellite and in-situ data Global Daily Sea Surface Temperature
MUR	Multi-scale Ultra-high Resolution
MW	Microwave
MY	Multi-year
NASA	National Aeronautics and Space Administration
NAVO	Naval Oceanographic Office
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OI	Optimal Interpolation
OSPO	Office of Satellite Products and Operations
OSTIA	Operational Sea Surface Temperature and Sea Ice Analysis
REP	Reprocessing
RSS	Remote Sensing System
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

Foundation sea surface temperature	**CF Definition**: The surface called "surface" means the lower boundary of the atmosphere. The sea surface foundation temperature is the water temperature that is not influenced by a thermally stratified layer of diurnal temperature variability (either by daytime warming or nocturnal cooling). The foundation temperature is named to indicate that it is the temperature from which the growth of the diurnal thermocline develops each day, noting that on some occasions with a deep mixed layer there is no clear foundation temperature in the surface layer. In general, sea surface foundation temperature will be similar

	to a night-time minimum or pre-dawn value at depths of between approximately 1 and 5 meters. In the absence of any diurnal signal, the foundation temperature is considered equivalent to the quantity with standard name sea_surface_subskin_temperature. The sea surface foundation temperature defines a level in the upper water column that varies in depth, space, and time depending on the local balance between thermal stratification and turbulent energy and is expected to change slowly over the course of a day. If possible, a data variable with the standard name sea_surface_foundation_temperature should be used with a scalar vertical coordinate variable to specify the depth of the foundation level. Sea surface foundation temperature is measured at the base of the diurnal thermocline or as close to the water surface as possible in the absence of thermal stratification. Only in situ contact thermometry is able to measure the sea surface foundation temperature. Analysis procedures must be used to estimate sea surface foundation temperature value from radiometric satellite measurements of the quantities with standard names sea_surface_skin_temperature and sea_surface_subskin_temperature. Sea surface foundation temperature provides a connection with the historical concept of a depth sea surface temperature considered representative of the oceanic mixed layer temperature that is typically represented by any sea temperature measurement within the upper ocean over a depth range of 1 to approximately 20 meters. The general term, depth sea surface temperature, has the standard name sea_surface_temperature with no associated vertical coordinate axis. Sea surface foundation temperature provides a more precise, well-defined quantity than depth sea surface temperature and, consequently, is more representative of the mixed layer temperature. The temperature of sea water at a particular depth (other than the foundation level) should be reported using the standard name sea_water_temperature and, wherever possible, supplying a vertical coordinate axis or scalar coordinate variable.
Level 4 SST analysis	A level 4 SST analysis refers to a globally complete, daily, gridded dataset of sea surface temperature. It is typically created by combining a background field with observations from satellite and in-situ platforms. A level 4 SST analysis uses some form of data assimilation technique to fille the gaps in observations to create a continuous, gap-free SST field. L4 SST analyses have been widely used for climate monitoring and in numeric weather prediction models.

RELEVANT DOCUMENT LIST

OSTIA QUID Document	CMEMS-SST-QUID-010-001[1]

[1] https://data.marine.copernicus.eu/product/SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001/description

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the product SST_GLO_PHY_L4_NRT_010_005, which is the Global Ocean Sea Surface Temperature Multi Product Ensemble. The product is created using the GHRSST (Group for High Resolution Sea Surface Temperature) Multi Product Ensemble (GMPE) system which runs daily at the Met Office in the United Kingdom (UK). The system uses global level 4 (L4) sea surface temperature (SST) analyses produced by various institutes around the world and produces a daily global ensemble median and standard deviation on a 0.25° x 0.25° regular latitude/longitude (lat/long) grid.

The GMPE product is mainly used as a monitoring tool for analysis data producers to assess the consistency between various SST analysis products. The standard deviation product can be used as an estimate of the confidence to be placed in each product contributing to the ensemble. It should be noted that the source of the GMPE median can vary from point to point, and therefore the GMPE median is not necessarily a consistent analysis (neither spatially nor temporally). Thus, this product is not suitable for uses such as boundary conditions to an atmosphere model or climate monitoring.

I.2 Summary of the results

The accuracy of the GMPE SST ensemble median has been assessed using a comparison to independent near surface Argo data[2] over the year 2013. The study used most recent Argo data at the time and similar study has not been repeated since 2014. Argo data has been chosen for this validation study as it is independent of all contributing analyses and Argo has also been shown to provide a very accurate suitable in-situ data source for SST validation studies as discussed further in section III. The GMPE median has a very low mean difference of -0.03 K globally relative to Argo data, with a standard deviation of the differences of 0.41 K, for the year 2013. There is some regional variation in the accuracy, with the South Atlantic having the largest standard deviation (0.54 K) and the Tropical Pacific region having the smallest standard deviation (0.24 K).

Martin et al. (2012) presented a similar validation study of the GMPE median compared with Argo data but for the year 2010. This study also compared the accuracy of the contributing SST analyses and found that the GMPE median provided a more accurate regional estimate of SST (based on the Argo comparisons) than any of the contributing analyses.

I.3 Estimated Accuracy Numbers

The Estimated Accuracy Numbers (EANs) obtained from the validation of the product are provided in Table 1.

Table 1. Mean (GMPE minus Argo) and standard deviation of the differences between GMPE median and near-surface Argo data during 2013 over the global ocean.

Variable	Units	Metrics	Values of metric
temperature	K	mean difference	-0.03
		standard deviation	0.41

[2] https://www.metoffice.gov.uk/hadobs/en4/

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production Centre

The Production Centre is the Met Office, the United Kingdom's national meteorological service.

II.2 Production subsystem

GHRSST Multi-Product Ensemble (GMPE) system reads in L4 analyses produced at various production centres and produces ensemble median of these analyses.

II.3 Product description

SST_GLO_PHY_L4_NRT_010_005 is produced daily at the Met Office using the GMPE system. It takes inputs from various SST L4 analysis production centres and produces ensemble median SST and standard deviation over the global fields. Each analysis which contributes to the GMPE product uses a different subset of observational data including both in situ and satellite SST data and different choices are made in analysis methods. Although all the contributing analyses are 'SST products', they are valid for different depths and time periods (within the day). For example, Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA) is a foundation SST product (i.e. the SST free of diurnal variability) so is valid immediately before sunrise. Other products are daily means in that they use data throughout the day, but this does not necessarily ensure that a daily mean is produced. Therefore, the GMPE ensemble median SST is not strictly valid at a particular depth or time during the day. In the validation presented here, the ability of the GMPE median SST to provide an accurate foundation SST is assessed.

Figure 1 shows a diagram detailing the GMPE processing chain at the Met Office. Further details of each stage are discussed below.

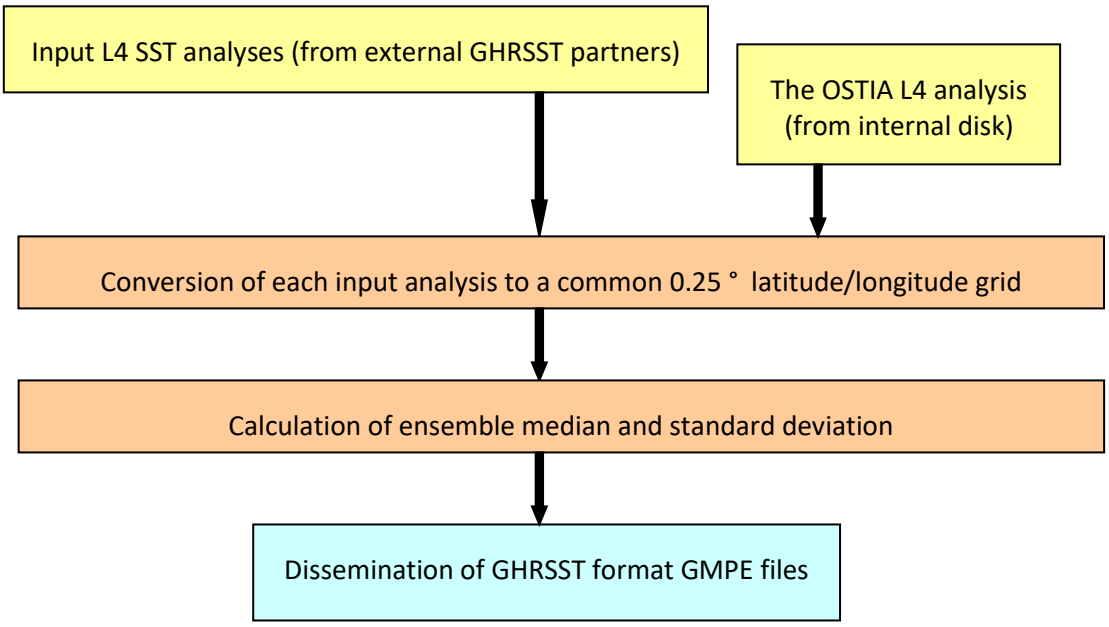

Figure 1. Schematic diagram of the GMPE processing chain at the Met Office.

II.3.1 Input analyses

L4 SST analyses that are currently contributing to the near-real time GMPE system are listed in Table. 2 alphabetically.

Table 2. List of L4 SST analyses contributing to the GMPE system

Product[3]	Production Centre	Reference
CMC 0.1°	Canadian Meteorological Center, Canada	Canada Meteorological Center, 2016; Brasnett, 2008
DMI	Danish Meteorological Institute, Denmark	DMI, 2007
FNMOC	Fleet Numerical Meteorology and Oceanography Centre, USA	Cummings, 2005; Cummings, 2011
GAMSSA	Bureau of Meteorology, Australia	Beggs et al., 2011; Zhong and Beggs, 2008
G1SST	NASA Jet Propulsion Laboratory, USA	JPL OurOcean Project, 2010; Chao et al., 2009
MGDSST	Japan Meteorological Agency, Japan	Kurihara et al., 2006
MUR	NASA Jet Propulsion Laboratory, USA	JPL MUR MEaSUREs Project, 2015
NAVO K10	Naval Oceanographic Office, USA	
OISST V2 AVHRR	National Oceanic and Atmospheric Administratio, USA	Reynolds et al., 2007
OSPO Night	NOAA Office of Satellite Products and Operations, USA	Office of Satellite Products and Operations, 2015
OSTIA	Met Office, UK	Donlon et al., 2012; CMEMS product, SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001
RSS MW	Remote Sensing Systems, USA	Gentemann et al., 2006
RSS MW/IR	Remote Sensing Systems, USA	Gentemann et al., 2006

It should be noted that the L4 SST analyses included in the GMPE can potentially vary daily due to data availability from the data providers. Full details of the contributing products are provided in the reference given.

II.3.2 Conversion of each input analysis to a common grid

The input L4 analyses are on different grids and are all re-gridded to a common 0.25 ° latitude/longitude grid using area averaging prior to the calculation of ensemble median.

[3] Abbreviations for SST products are used as the name of the product, the full name of abbreviated institutes or products are spelled in full form in the abbreviation table where applicable.

II.3.3 Calculation of ensembles median and standard deviation

At each of the 0.25 ° latitude/longitude grid point, the median and standard deviation of the input SST values are calculated, and the results are saved to a netCDF file format.

II.3.4 Additional fields

As well as the above fields, a number of additional fields are provided:

1. The number of analyses contributing to the ensemble at each grid point.
2. The identifier of the analysis which provides the ensemble median at each grid point.
3. The anomaly of each analysis from the median value at each grid point.
4. The horizontal gradients of each contributing analysis, and of the GMPE median. These are the averaged north-south/east-west gradients calculated using centred differencing on the original input analysis grid (i.e. before interpolation to the 0.25 ° grid).

III VALIDATION FRAMEWORK

The accuracy of GMPE median is assessed by comparison to near-surface Argo SST measurements. In this section, a general description of the framework is given, and this has remained unchanged since the last assessment in 2013. The L4 SST analyses used in the assessment in 2013 are slightly different to those shown in Table 2. Specifically, the two microwave products RSS MW and RSS MW/IR were not available for the test period of 2013. DMI, G1SST, MUR and OSPO_Night products were also not available in 2013. CMC updated their product to CMC 0.1 ° and the newer version replaced the coarser version in 2017. Real-time, global SST (RTG SST) product discontinued in 2020. Basic information of the SST analyses used in the 2013 assessment are as the following:

- OSTIA, Met Office, UK (Donlon et al., 2012; CMEMS product, SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001)
- CMC, Canadian Meteorological Center, Canada (Brasnett, 2008; Brasnett, 1997)
- FNMOC, Fleet Numerical Meteorology and Oceanography Centre, USA (Cummings, 2005; Cummings, 2011.
- GAMSSA, Bureau of Meteorology, Australia (Beggs et al., 2011; Zhong and Beggs, 2008.)
- MGDSST, Japan Meteorological Agency, Japan (Kurihara et al. ,2006)
- NAVO K10, Naval Oceanographic Office, USA
- OISST.v2 AVHRR, National Oceanic and Atmospheric Administration, USA (Reynolds et al., 2007)
- RTG, National Centers for Environmental Prediction, USA (Gemmill et al., 2007)

For the GMPE median assessment, the shallowest Argo observations available between 3 and 5 m depth were used. Argo data are chosen for this validation, rather than drifting buoy data, as no Argo data have been used in the creation of any of the contributing analyses and thus provide an independent data source to assess the accuracy of the GMPE median. Argo data have a high level of accuracy with an uncertainty value of 0.005 K (Oka and Ando, 2004) and although the Argo spatial data coverage is lower than for drifting buoys, Argo floats are now present in most of the world's oceans. Figure 2 shows the spatial distribution of Argo floats for an example month (April 2013) during the validation period. Argo data have also been shown to agree well with drifting buoy data (e.g. Figure 1 in Fiedler et al., 2014). Near surface Argo data have previously been used to validate SST products (e.g. Roberts-Jones et al., 2012; Corlett et al., 2014; Fiedler et al., 2014). In particular, Martin et al. (2012) assessed the accuracy of the GMPE median by comparison to Argo data for the year 2010.

It is worth noting that although Argo was independent to all participating L4 SST products in the 2013 validation, the current operational GMPE median is not completely independent to Argo. This is because one of the participating products, MUR, ingests Argo in the production chain; hence, since the inclusion of MUR in GMPE, GMPE median was no longer completely independent to Argo data. However, MUR is the only L4 product that ingests Argo and, the number of Argo observations is much smaller than the other observation types used in the product. Argo data are still used as the routine assessment tool for GMPE accuracy.

The Argo data used here were obtained from the EN4 data-set[4]; Good et al., 2013). The GMPE median was bi-linearly interpolated to the Argo data location on each day. The mean and standard deviation of

[4] www.metoffice.gov.uk/hadobs/en4

the Argo minus GMPE differences were then calculated for the whole globe and over various ocean regions (as defined in Figure 3).

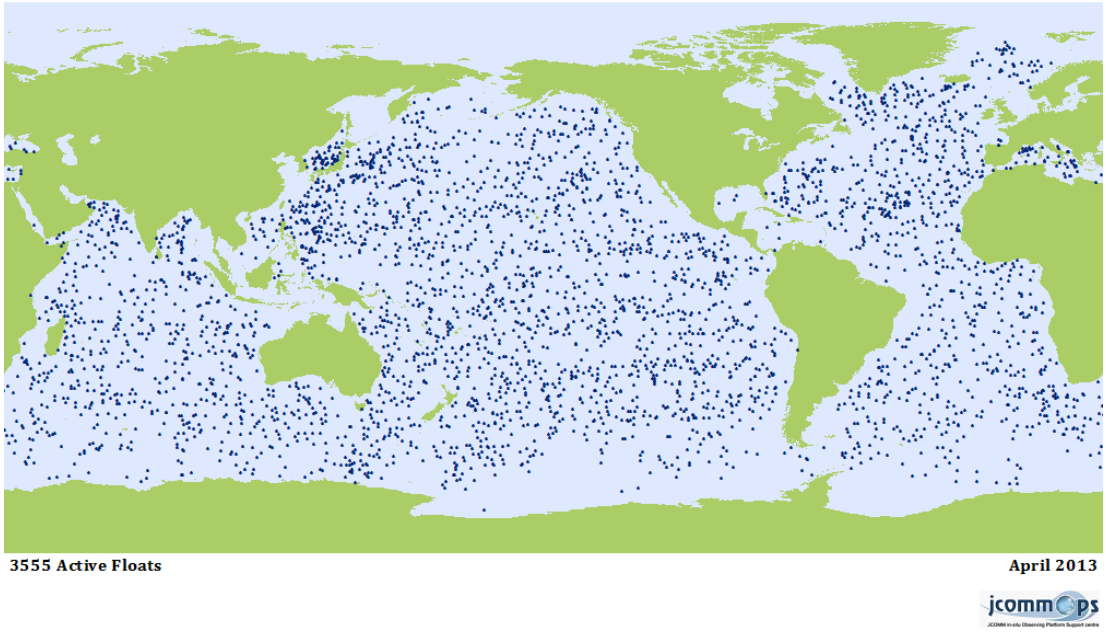

Figure 2. Example global spatial distribution for Argo floats for April 2013[5]).

[5] http://wo.jcommops.org/cgi-bin/WebObjects/Argo/

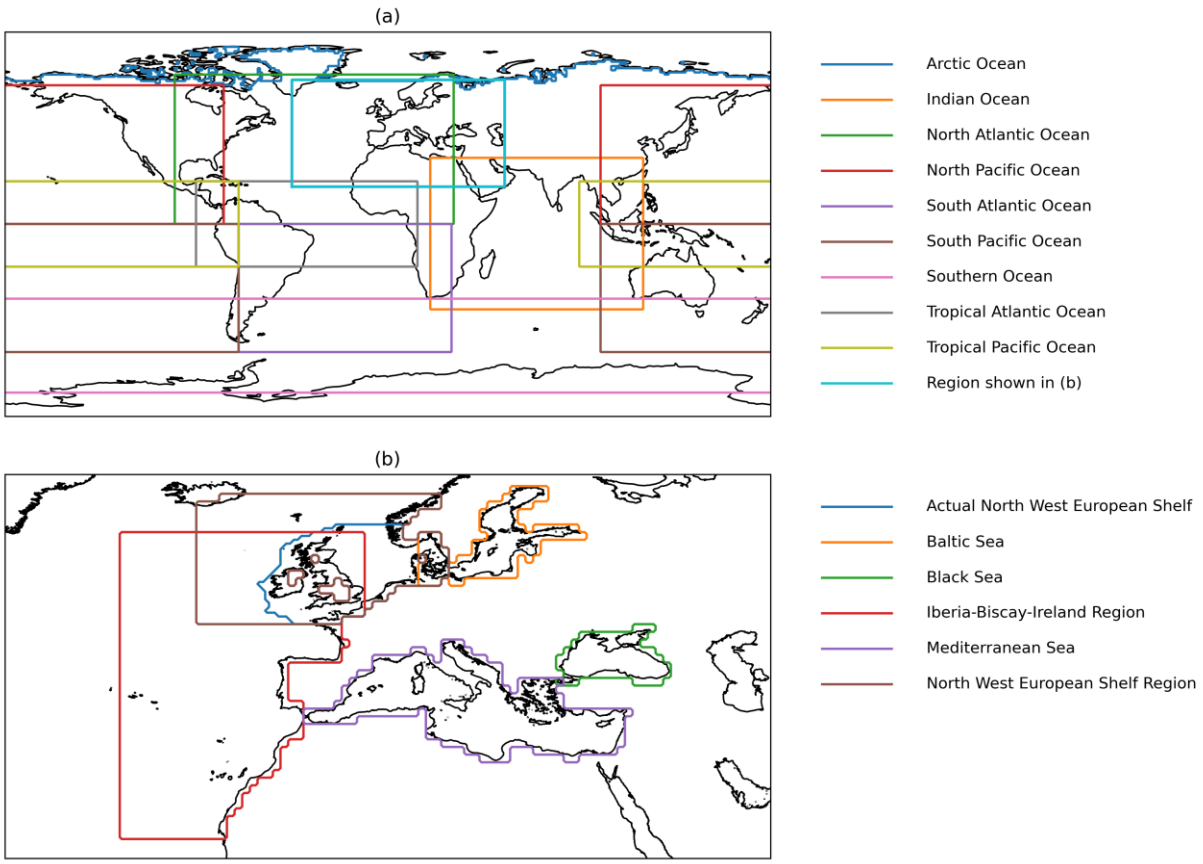

Figure 3. (a) Geographical regions used in the GMPE median validation, except for the European regional areas within the cyan box. (b) Zoomed in view of the European region showing the GMPE validation regions within that area.

IV VALIDATION RESULTS

Table 3 details the mean and standard deviation of the “GMPE median minus Argo" differences for the whole globe and for individual ocean regions. Regions with an insufficient number of Argo observations for the validation are not shown in Table 3.

Table 3. Global and regional mean (GMPE minus Argo) and standard deviation differences between GMPE median and near-surface Argo data during 2013.

Region	Mean difference (K)	Standard deviation of differences (K)	Total number of observations
Global	-0.03	0.41	87579
Mediterranean Sea	0.06	0.34	1768
North Atlantic	-0.05	0.41	15289
Tropical Atlantic	-0.08	0.29	6155
South Atlantic	-0.07	0.54	7782
North Pacific	-0.04	0.44	27306
Tropical Pacific	-0.02	0.24	23771
South Pacific	-0.03	0.31	22575
Indian Ocean	-0.03	0.34	14960
Southern Ocean	-0.05	0.50	16633

The comparison shows that the GMPE median has a very small global bias, with a standard deviation of 0.41 K. There are regional differences with the Tropical Atlantic and Tropical Pacific having greater accuracy and the South Atlantic and Southern Ocean having lower accuracies. A previous work reported that the standard deviation calculated using Argo observations recorded between January 2007 and December 2013 are 0.3 K, 0.29 K, 0.52 K and 0.54 K for Tropical Atlantic, Tropical Pacific, South Atlantic and Southern Ocean. Fiedler, et al., .

The same Argo comparison has been carried out for the near-real time OSTIA product (SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001), which can then be compared with the GMPE median comparison (Note that the OSTIA product is one of the analyses used in GMPE). Results are shown in Table 4. Comparison of tables 3 and 4 shows that the standard deviation of the differences for the GMPE median is smaller than the contributing OSTIA product both globally and for all the individual regions.

Table 4. Global and regional mean (OSTIA minus Argo) and standard deviation differences between OSTIA and near-surface Argo data during 2013.

Region	Mean difference (K)	Standard deviation of differences (K)	Total number of observations
Global	-0.06	0.46	87947
Mediterranean Sea	0.01	0.35	1768
North Atlantic	-0.09	0.45	15299
Tropical Atlantic	-0.12	0.33	6155
South Atlantic	-0.11	0.59	7797
North Pacific	-0.05	0.50	27329
Tropical Pacific	0.00	0.29	23771
South Pacific	-0.03	0.37	22581
Indian Ocean	-0.02	0.36	14960
Southern Ocean	-0.17	0.55	16931

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section describes major changes that may have impacted the quality of the operational product since the 2013 assessment. Table 5 lists the past noticeable changes and more detailed assessment of its impact are presented later in this section. Please note that, the product was removed from Copernicus Marine Service's data catalogue between January 2023 and December 2025, following the UK's exit from the European Union. The product will be available again in the catalogue from March 2026.

Table 5. Major changes impacting the quality of the product.

Date	Change/Event description	System version	Other
16/10/2017	Inclusion of additional analyses	OS40	
31/03/2026	Inclusion of additional analyses	OS47	

On October 16th 2017 the GMPE ensemble was expanded to include new analyses. The impact of the new analyses on the GMPE median was investigated by running a trial for 28 days, and examining the differences between the two GMPE median products (current GMPE median and GMPE median produced using current and new analyses). Note that during this testing phase an additional OSPO product (OSPO all) was included but was excluded from the final ensemble due to the similarity with OSPO night; additionally, not all the analyses are available currently. The average difference between the two GMPE median products, as well as the standard deviation and maximum difference are shown in Figure 4.

Examination of the daily fields indicates that the magnitude of change between the new GMPE median and the baseline GMPE median generally ranges from -0.3 to +0.5 Kelvin, with some small areas showing up to 1.0 Kelvin warmer SSTs (relative to the baseline GMPE median). A general pattern is evident from the daily differences, with the new GMPE median (current and new analyses) being generally up to 0.3 Kelvin cooler than the baseline GMPE median in many broad regions, but with distinct and consistent warmer SSTs (relative to the baseline) in the northeast Pacific, Gulf Stream, Kuroshio Current, and the Agulhas Current.

These patterns are evident in the average difference between the two GMPE medians (Figure 4), indicating that the inclusion of the new analyses did not have a large impact on the GMPE median. SSTs were on average up to 0.1 Kelvin cooler (relative to the baseline GMPE median) in regions of the south Pacific, coastal South America, North Atlantic, and coastal central Africa (Figure 5). Warmer SSTs (relative to the baseline) were evident around Indonesia (up to 0.4 Kelvin warmer), Agulhas Current (up to 0.5 Kelvin warmer), the Gulf Stream and Kuroshio Current (up to 1.0 Kelvin warmer). The Arctic showed the largest differences, with an average difference of up to 1.5 Kelvin warmer or cooler than the baseline in some areas. These areas (Arctic, Gulf Stream, and Kuroshio) also showed the greatest variability (Figure 4).

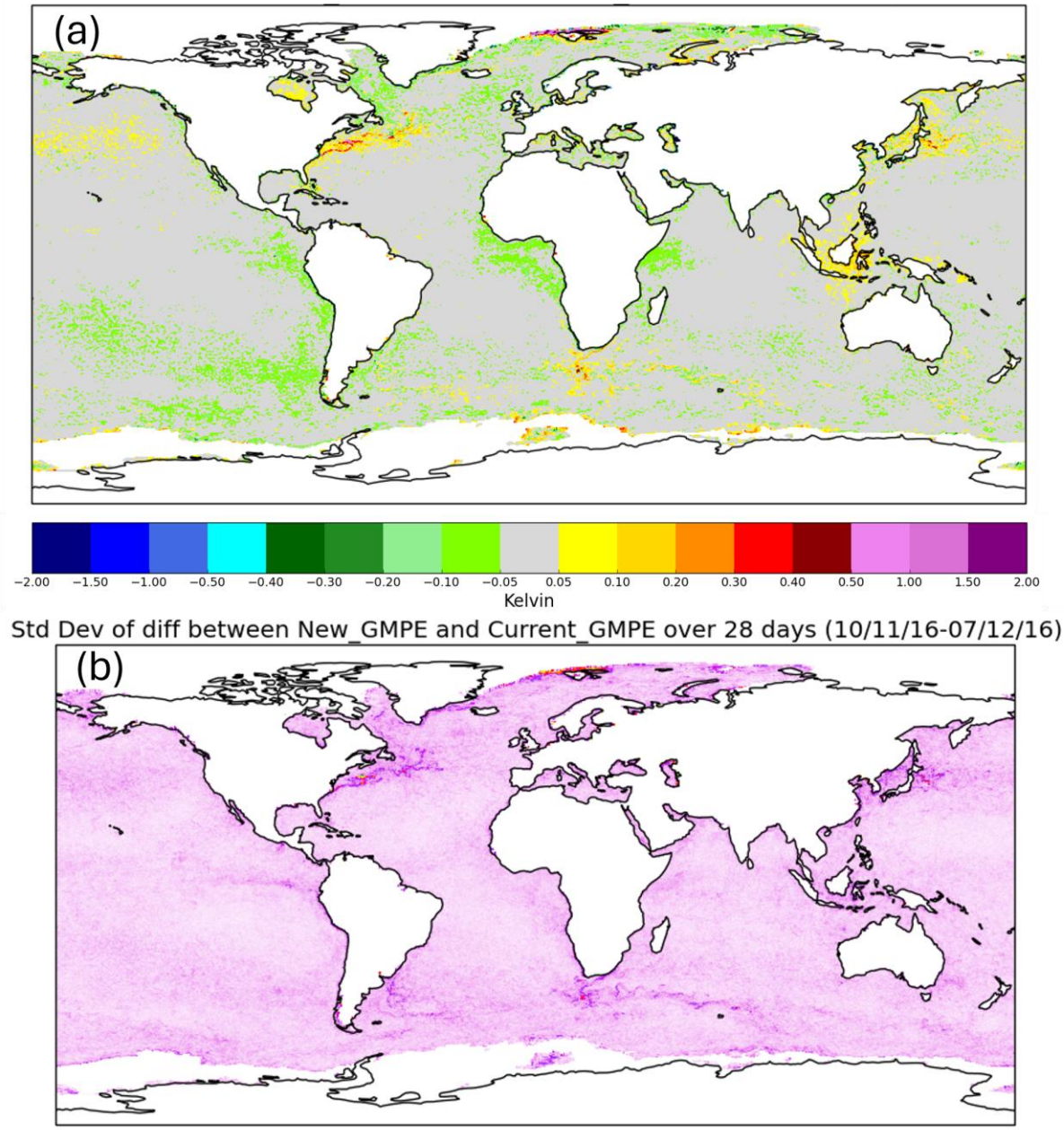

Figure 4. (a): Mean difference between the GMPE median with new analyses and the current GMPE median over a 28-day period (note the non-uniform scale bar). Positive values mean that adding the new analyses to the GMPE median resulted in, on average, warmer SSTs compared to using only the current analyses, while negative values indicate cooler SSTs. The colours show how much including a new analysis in the GMPE product affects the outcome compared to the control. Please be aware that the scale bar is not uniform. (b): Standard deviation of the difference between the new and the current GMPE medians.

Each individual analysis and the GMPE median were assessed using Argo data using the analyses for the same 28-day period used above. Results are shown in Figure 5. The new analyses are similar to the existing analyses in their mean standard deviation compared to Argo, with the exception that G1SST has the opposite sign mean difference to Argo. However, this could be a statistical anomaly.

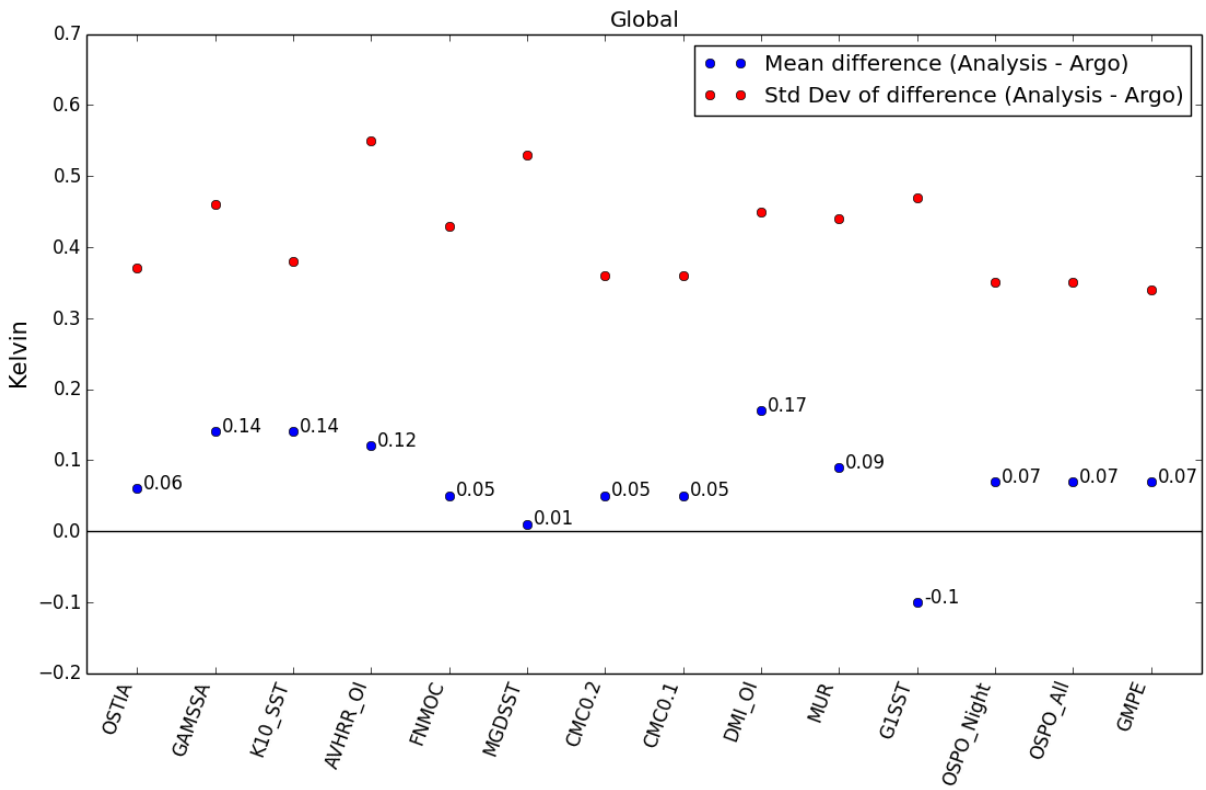

Figure 5. Global performance of the SST analyses (n=13) and the GMPE median measured against Argo observations over a 28-day period (10th November 2016 - 7th December 2016 inclusive), showing mean difference (Kelvin) between the analysis and Argo[6] (Analysis minus Argo; blue circles) and standard deviation (red circles).

[6] https://www.metoffice.gov.uk/hadobs/en4/

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current versions.

There have not been major changes to GMPE since the inclusion of additional analyses in 2017. This was the same GMPE system as the one when GMPE rejoined CMEMS in June 2025.

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